

Arithmetic Operators

1. A product costs ₹450 and a customer buys 6 of them. Calculate the total bill.

price = 450

quantity = 6

total_bill = price * quantity

print("Total bill:", total_bill)

OUTPUT: Total bill: 2700

2. A student scored 480 marks out of 600. Calculate the percentage.

marks = 480

total_marks = 600

percentage = (marks / total_marks) * 100

print("Percentage:", percentage)

OUTPUT: Percentage: 80.0

3. A company has 125 employees. Divide them equally into 8 teams. Find how many employees are in each team.

employees = 125

teams = 8

employees_per_team = employees // teams

print("Employees in each team:", employees_per_team)

OUTPUT: Employees in each team: 15

4. A bakery made 95 cupcakes and packs them into boxes of 12. Find how many completely filled boxes can be made.

cupcakes = 95

box_size = 12

filled_boxes = cupcakes // box_size

print("Completely filled boxes:", filled_boxes)

OUTPUT: Completely filled boxes: 7

5. After packing the cupcakes into boxes of 12, find how many cupcakes are left.

```
cupcakes = 95
box_size = 12

remaining_cupcakes = cupcakes % box_size

print("Remaining cupcakes:", remaining_cupcakes)
```

OUTPUT: Remaining cupcakes: 11

6. A square has a side length of 14 cm. Find its area.

```
side = 14

area = side * side

print("Area of square:", area)
```

OUTPUT: Area of square: 196

7. A number is multiplied by itself 5 times. Write a program for the calculation.

```
number = 3

result = number ** 5

print("Result:", result)
```

OUTPUT: Result: 243

8. A person earns ₹1,250 per day for 28 days. Find the total earnings.

```
daily_earning = 1250
days = 28

total_earnings = daily_earning * days

print("Total earnings:", total_earnings)
```

OUTPUT: Total earnings: 35000

9. Rahul has ₹2,500 and spends ₹1,375. Find the remaining balance.

```
balance = 2500
```

```
spent = 1375
```

```
remaining_balance = balance - spent
```

```
print("Remaining balance:", remaining_balance)
```

OUTPUT: Remaining balance: 1125

10. Find the average of three numbers: 45, 78, and 92.

```
num1 = 45
```

```
num2 = 78
```

```
num3 = 92
```

```
average = (num1 + num2 + num3) / 3
```

```
print("Average:", average)
```

OUTPUT: Average: 71.66666666666667

Assignment Operators

11. Store the value 500 in a variable called `balance`.

```
balance = 500
```

```
print("Balance:", balance)
```

OUTPUT: Balance: 500

12. A wallet initially has ₹800. Add ₹250 to it using a shortcut assignment.

```
wallet = 800
```

```
wallet += 250
```

```
print("Wallet balance:", wallet)
```

OUTPUT: Wallet balance: 1050

13. A player has 100 health points. Reduce the health by 35 using a shortcut assignment.

```
health = 100
```

```
health -= 35
```

```
print("Remaining health:", health)
```

OUTPUT: Remaining health: 65

14. A salary is ₹40,000. Increase it by multiplying it by 2 using a shortcut assignment.

```
salary = 40000
```

```
salary *= 2
```

```
print("Updated salary:", salary)
```

OUTPUT: Updated salary: 80000

15. A total distance is 600 km. Divide it equally among 3 drivers using a shortcut assignment.

```
distance = 600
```

```
distance /= 3
```

```
print("Distance for each driver:", distance)
```

OUTPUT: Distance for each driver: 200.0

16. A number is 96. Update it by dividing it into groups of 5 and keeping only the complete groups.

```
number = 96
```

```
number //= 5
```

```
print("Complete groups:", number)
```

OUTPUT: Complete groups: 19

17. Store the value 8 in a variable and update it by raising it to the power of 3 using a shortcut assignment.

```
number = 8
```

```
number **= 3
```

```
print("Result:", number)
```

OUTPUT: Result: 512

18. A warehouse has 125 items. After packing them into groups of 9, update the variable to store only the remaining items.

```
items = 125
```

```
items %= 9
```

```
print("Remaining items:", items)
```

OUTPUT: Remaining items: 8

Comparison / Relational Operators

19. Check whether Rahul's age (18) is greater than Ramesh's age (16).

```
rahul_age = 18  
ramesh_age = 16
```

```
result = rahul_age > ramesh_age
```

```
print("Rahul is older than Ramesh:", result)
```

OUTPUT: Rahul is older than Ramesh: True

20. Check whether the temperature is less than 0°C.

```
temperature = -5
```

```
result = temperature < 0
```

```
print("Temperature is below 0°C:", result)
```

OUTPUT: Temperature is below 0°C: True

21. Check whether a student scored at least 35 marks.

```
marks = 42
```

```
result = marks >= 35
```

```
print("Student passed:", result)
```

OUTPUT: Student passed: True

22. Check whether a product price is not equal to ₹999.

```
price = 850
```

```
result = price != 999
```

```
print("Price is not equal to ₹999:", result)
```

OUTPUT: Price is not equal to ₹999: True

23. Check whether two passwords entered by a user are exactly the same.

```
password1 = "Python123"  
password2 = "Python123"
```

```
result = password1 == password2
```

```
print("Passwords are the same:", result)
```

OUTPUT: Passwords are the same: True

24. Check whether a person's height is greater than or equal to 170 cm.

```
height = 175
```

```
result = height >= 170
```

```
print("Height is greater than or equal to 170 cm:", result)
```

OUTPUT: Height is greater than or equal to 170 cm: True

25. Check whether the stock quantity is less than or equal to 10.

```
stock = 8
```

```
result = stock <= 10
```

```
print("Stock is less than or equal to 10:", result)
```

OUTPUT: Stock is less than or equal to 10: True

26. Check whether the entered PIN is different from the stored PIN.

```
stored_pin = 1234
```

```
entered_pin = 5678
```

```
result = entered_pin != stored_pin
```

```
print("PIN is different:", result)
```

OUTPUT: PIN is different: True

Logical Operators

27. A student passes only if marks are above 35 and attendance is at least 75%. Write the condition.

```
marks = 40
```

```
attendance = 80
```

```
result = marks > 35 and attendance >= 75
```

```
print("Student Passed:", result)
```

OUTPUT: Student Passed: True

28. A customer gets free delivery if they are a premium member or their purchase is above ₹1,000.

```
premium_member = False
```

```
purchase_amount = 1500
```

```
result = premium_member or purchase_amount > 1000
```

```
print("Free Delivery:", result)
```

OUTPUT: Free Delivery: True

29. A website allows login only if the username and password are both correct.

```
username_correct = True
```

```
password_correct = True
```

```
result = username_correct and password_correct
```

```
print("Login Successful:", result)
```

OUTPUT: Login Successful: True

30. A user can enter the examination hall if they have an ID card or a hall ticket.

```
id_card = False
```

```
hall_ticket = True
```

```
result = id_card or hall_ticket
```

```
print("Allowed to Enter:", result)
```

OUTPUT: Allowed to Enter: True

31. Check whether a number is not negative.

```
number = 15
```

```
result = not (number < 0)
```

```
print("Number is not negative:", result)
```

OUTPUT: Number is not negative: True

32. A person is eligible for a driving license only if they are at least 18 years old and have passed the driving test.

```
age = 20
```

```
passed_test = True
```

```
result = age >= 18 and passed_test

print("Eligible for Driving License:", result)

OUTPUT: Eligible for Driving License: True
```

33. A movie ticket discount is available if the customer is a student or a senior citizen.

```
student = False
senior_citizen = True

result = student or senior_citizen

print("Discount Available:", result)

OUTPUT: Discount Available: True
```

34. A machine starts only if the emergency switch is not turned on.

```
emergency_switch = False

result = not emergency_switch

print("Machine Starts:", result)

OUTPUT: Machine Starts: True
```

Bitwise Operators

35. Find the binary result after performing a bitwise operation between 12 and 10.

```
num1 = 12
num2 = 10

result = num1 & num2

print("Bitwise AND:", result)
print("Binary:", bin(result))
```

OUTPUT: Bitwise AND: 8
Binary: 0b1000

36. Find the binary result after performing a bitwise operation between 9 and 5.

```
num1 = 9
num2 = 5

result = num1 | num2

print("Bitwise OR:", result)
print("Binary:", bin(result))
```

OUTPUT: Bitwise OR: 13
Binary: 0b1101

37. Find the binary result after performing a bitwise operation between 15 and 6.

```
num1 = 15
num2 = 6

result = num1 ^ num2

print("Bitwise XOR:", result)
print("Binary:", bin(result))
```

OUTPUT: Bitwise XOR: 9
Binary: 0b1001

38. Shift the binary representation of 20 by 2 positions toward the left.

```
number = 20
```

```
result = number << 2
```

```
print("Left Shift:", result)  
print("Binary:", bin(result))
```

OUTPUT: Left Shift: 80
Binary: 0b1010000

39. Shift the binary representation of 64 by 3 positions toward the right.

```
number = 64
```

```
result = number >> 3
```

```
print("Right Shift:", result)  
print("Binary:", bin(result))
```

OUTPUT: Right Shift: 8
Binary: 0b1000

40. Find the bitwise complement of the number 18.

```
number = 18
```

```
result = ~number
```

```
print("Bitwise Complement:", result)
```

OUTPUT: Bitwise Complement: -19